MIL-L-51450A(EA)
30 April 1985
SUPERSEDING
MIL-L-51450(EA)
11 October 1978

MILITARY SPECIFICATION

LAUNCHER, PROJECTILE, 64 MM - M234

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of launcher for the M742 and M743 riot control projectiles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-T-97 - Tape, Pressure Sentitive Adhesive, Filament Reinforced.

MILITARY

MIL-B-2427 - Boxes, Ammunition Packing; Wood, Nailed.
MIL-B-7883 - Brazing of Steel, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 122-3-30 - Launcher, Projectile, 64 mm - M234.

PRODUCT ASSURANCE TEST PROCEDURES

136-400-100 - Inspection Standard for Evaluation of Brazed Joints and Structural Components for Projectile Launching Cup, M234 Launcher.

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM 9-1010-224-10 - Launcher, Projectile, 64 MM: Riot Control, M234.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 122-3-30 and on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the launcher shall conform to the specifications and drawings listed on Drawing 122-3-30.

3.2 Assembly. The launcher shall be assembled as shown on Drawing 122-3-30.

3.3 Performance.

3.3.1 Functioning, velocity and dispersion. When tested in accordance with 4.4.4.1, the launcher shall show no sign of component failure. It shall eject a projectile with a velocity of 170-200 feet per second (ft/sec) and each projectile shall be within 18 inches of the center of impact when fired at a distance of 40 meters.

3.3.2 Brazing and structural adequacy. The brazing and structural adequacy of the projectile launching cup shall conform to the requirements specified in MIL-B-7883. The radiographic requirements of MIL-B-7883 shall be supplemented by Inspection Standard No. 136-400-100 when tested in accordance with 4.4.4.2.

3.3.3 Reliability. The launcher shall have a minimum mean rounds between failure of 170 when tested in accordance with 4.4.4.3.

3.3.4 Proof-firing. All projectile launching cups (Drawing 122-3-45) shall not experience metal parts failure, nor show visual evidence of any movement or cracking of a braze joint or have any loose or cracked keys or supports when proof-fired in accordance with 4.4.4.4. Accepted cups shall be proof-marked in accordance with Drawing 122-3-45.

3.4 Preproduction. Prior to the start of regular production, a preproduction lot of components shall be produced in accordance with this specification for examination and tests (see 4.3).

3.5 Workmanship. The launcher, including components, shall be free from burrs, sharp edges, foreign matter and damages such as cracks or deformation which could impair its intended use. The recoil mechanism buffer (Drawing 122-3-140) shall be free from imperfections listed in table I of MIL-STD-407 and damage, such as rips and tears, which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction and reliability qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of components suitable to assemble 100 launchers and the level A preparation for delivery shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. A sample of components, subassemblies and assemblies for 5 launchers shall be examined 100 percent to all drawing dimensions and requirements.

4.3.2.2 For tests.

4.3.2.2.1 Functioning, velocity and dispersion. Assemble 10 launchers from the preproduction lot and test in accordance with 4.4.4.1.

4.3.2.2.2 Brazing and structural adequacy. Five (5) projectile launching cups shall be examined and tested for hardness and five (5) projectile launching cups shall be examined and tensile stress-tested in accordance with 4.4.4.2.

4.3.2.2.3 Reliability qualification. Assemble three (3) launchers from the preproduction lot and test in accordance with 4.4.4.3.

4.3.2.3 Proof-firing. Each projectile launching cup shall be proof-fired in accordance with 4.4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction lot shall comply with the examination and tests specified in 4.3.2 to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the launchers produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of each component from one manufacturer shall be contained in any one lot of assembled launchers.

4.4.2 Sampling.

4.4.2.1 For examination and test.

4.4.2.1.1 Examination, functioning, velocity and dispersion. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.1.2 Brazing and structural adequacy. Sampling shall be conducted in accordance with Inspection Standard No. 136-400-100.

4.4.2.2 Proof-firing. There shall be no sampling. Each projectile launching cup shall be proof-fired in accordance with 4.4.4.4.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample launcher components and the level A preparation for delivery shall be examined in accordance with the classification of defects (4.4.3.4).

4.4.3.2 For test.

4.4.3.2.1 Functioning, velocity and dispersion. The sample launchers shall be functioned in accordance with 4.4.4.1. The AQL shall be 2.5 percent defective.

4.4.3.2.2 Brazing and structural adequacy. The sample projectile launching cups shall be examined and stress-tested in accordance with 4.4.4.2.

4.4.3.3 For critical defects. Each launcher in the lot shall be examined for its critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Launcher, projectile, 64 mm - M234 (Drawing 122-3-30).

Categories	Defects
Critical:	
1	Deflector missing
2	Manifold retaining pin not in manifold groove
3	Buffer missing or improperly assembled

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Major: 100% Inspection

101	Cup proof-mark missing

AQL 1.0 percent defective

102	Assembled improperly
103	Components missing or incorrect
104	Workmanship (3.5)

(b) Cup, projectile launching (Drawing 122-3-45).

Categories	Defects
Critical:	None defined
Major:	100% Inspection
101	Proof-mark missing or illegible

AQL 1.0 percent defective

102	Run-outs not within tolerance
103	Brazing incorrect
104	Assembled incorrectly
105	Rifling keys incorrectly located or loose
106	Key support loose or gap between support and rifling key
107	Major diameter or length of thread incorrect
108	Failure to meet ring gage requirement
109	Hardness incorrect
110	Diameter over keys incorrect
111	Weight incorrect
112	Workmanship (3.5)

(c) Sleeve, threaded (Drawing 122-3-24).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread incorrect
102	Length incorrect
103	Radius on corner of shoulder incorrect

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(d) Cup (Drawing 122-3-139).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Inside diameter underbead incorrect
102	Outside diameter overbead incorrect
103	2.089 diameter incorrect
104	Run-outs not within tolerance
105	Cone angles incorrect
106	Wall thickness incorrect

(e) Lever, lock release (Drawing 122-3-17).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	.430 radius incorrect
102	.320 radius incorrect
103	Hardness incorrect
104	Workmanship (3.5)

(f) Deflector (Drawing 122-3-16).

Categories	Defects
Critical:	
1	Hardness incorrect
Major:	AQL 1.0 percent defective
101	Overall diameter incorrect
102	Positions not within tolerance
103	Overall length incorrect
104	0.500 diameter incorrect

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(g) Manifold, projectile (Drawing 122-3-33).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Locking groove incorrect
102	Locking groove incorrectly located
103	Center hole diameter incorrect
104	Overall outside diameter incorrect
105	Diameter of shoulder incorrect
106	Hardness incorrect
107	1.799 diameter incorrect
108	Run-out of 1.824 diameter not within tolerance
109	Workmanship (3.5)

(h) Plate, retaining launcher cup (Drawing 122-3-27).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread incorrect
102	Outside diameter incorrect
103	Outside diameter run-out incorrect
104	Face run-out incorrect
105	Workmanship (3.5)

(i) Sleeve, lock release (Drawing 122-3-39).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	.870 diameter incorrect
102	Threads incorrect
103	Hardness incorrect
104	Workmanship (3.5)

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(j) Buffer, recoil mechanism assy (Drawing 122-3-29).

Categories	Defects
Critical:	
1	Buffer retainer missing or incorrectly assembled

(k) Launcher, body-inseparable assembly (Drawing 122-3-31).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Assembly orientation incorrect
102	Gap at shoulder of bonded parts

(l) Ring, retaining (Drawing 122-3-34).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Wire diameter incorrect

(m) Housing, buffer (Drawing 122-3-47).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	1.800 diameter incorrect
102	.281 slot position incorrect
103	.312 diameter position incorrect
104	.094 diameter hole location incorrect

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(n) Barrel (Drawing 122-3-51).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	.058 groove location and size incorrect
102	1.806 diameter incorrect
103	.4975 diameter incorrect
104	.256 location incorrect
105	2.685 diameter or run-out incorrect

(o) Arm (Drawing 122-3-76).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	.094 diameter hole location incorrect
102	.340 width location incorrect

(p) Buffer, recoil mechanism (Drawing 122-3-140).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Durometer incorrect
102	Length incorrect
103	1.312 diameter incorrect
104	.500 diameter or position not within tolerance
105	1.00 diameter or position not within tolerance
106	Workmanship (3.5)

(q) Retainer (Drawing 122-3-132).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Positions not within tolerance

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102	.249 diameter incorrect
103	.062 diameter incorrect
104	Workmanship (3.5)

(r) Detent (Drawing 122-3-133).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Positions not within tolerance
102	.230 diameter incorrect
103	Spherical radius incorrect
104	Workmanship (3.5)

(s) Retainer, buffer (Drawing 122-3-28).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Position not within tolerance
102	Diameters incorrect
103	.06 radius incorrect

(t) Packaging (section 5) (Drawing 122-3-69).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Cushion insert broken
102	Hardware improperly engaged
103	Marking incorrect or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning, velocity and dispersion. An indoor firing range shall be used. The launcher shall be assembled to an M16A1/AR15 rifle, loaded, fired and serviced in accordance with TM 9-1010-224-10. Five rounds shall be fired.

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(a) The velocity of each round shall be measured using photo-electric ballistic screens and elapsed time counters. The distance from the center of the screens to the launcher barrel shall be 5.0 ± 0.1 ft. Record each velocity.

(b) Measure the dispersion distance from the center of impact for each round. A launcher fails if 1 or more out of 5 rounds does not meet the requirement.

(c) Remove the launcher cup assembly (Drawing 122-3-45), inspect for failure, clean, lubricate and reassemble. Failure of any component is cause for rejection of the launcher.

4.4.4.2 Brazing and structural adequacy. Test and examine in accordance with Inspection Standard No. 136-400-100.

4.4.4.3 Reliability. The launcher shall be assembled to a M16A1/AR15 rifle, loaded, fired and serviced in accordance with TM 9-1010-224-10. All rounds shall be shoulder fired over a rest. The firing sequence shall be one round every 30 seconds for 6 rounds, then a 3 minute cooling down period. Repeat for a total of 30 rounds. Disassemble, clean, inspect, lubricate, and reassemble the launcher and repeat in 30 round cycles. Velocities shall be measured (see 4.4.4.1) and recorded. Projectile impact data shall be recorded to the nearest centimeter for the 1st, 4th, 7th, etc. 30-round groups for information. Projectiles for these groups shall be unfired. 30-round groups 2, 3, 5, 6, 8, etc. shall be fired into a net at 35 meters and re-used provided the breakbands remain intact. Fire each launcher until all three launchers experience one or more of the listed failures or an average of 170 rounds per launcher is obtained, whichever occurs first. The total number of rounds until failure is noted and the nature of the failure shall be recorded for each launcher. These launchers shall be considered destroyed in testing and no components may be reused for production (see 6.1).

Failures

(a) Key wear to edge of center rivet head

(b) Any crack, dent, or distortion (visual or tactile)

(c) Loss of any component

(d) Loosening of any riveted, pinned, press fit or bonded components

(e) Detent fails to retain cup during loading and handling

(f) Binding of sleeve, latches or cup

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NOTE: Cup failures during firing may be evidenced by a sudden change in velocity or an erratic projectile flight. If there is no visible evidence of failure the next round should be fired (using an unfired projectile). If the condition persists disassemble the launcher and identify the cause. If not attributable to the launcher, reassemble and continue the test.

4.4.4.4 Proof-firing. Each projectile launching cup shall be assembled into an M234 launcher which shall be assembled to an M16A1/AR15 rifle, loaded and fired in accordance with TM 9-1010-224-10. Each launcher cup shall be fired once, cleaned and examined. Projectiles shall be fired into a net at 35 meters for soft recovery and re-used provided the breakbands remain intact. Proof-mark accepted cups.

5. PACKAGING

5.1 Packaging, level A. Each launcher shall be individually packaged in a pre-formed polystyrene container conforming to Drawing 122-3-72. The container shall be closed using tape, one-inch wide, type II, class B of PPP-T-97. The tape shall be applied to the container as shown on Drawing 122-3-69.

5.2 Packing, level A. Each launcher, packaged as specified in 5.1, shall be packed in a wood shipping container conforming to Drawing 122-3-69. The shipping container shall be closed and strapped in accordance with the closure requirements of MIL-B-2427, using flat galvanized steel strapping conforming to type I or II (heavy duty), class A, style optional, 3/8 by 0.020 inch of QQ-S-781.

5.3 Marking. Unless otherwise specified in the contract or order, shipping containers shall be marked in accordance with Drawing 122-3-69.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level A packaging required.
(c) Preproduction:

(1) Time allowed for contractor's submission of samples for Government test and evaluation after award of contract,
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.
(4) Disposition instructions for the reliability test launchers.

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6.2 Caution.

(a) The M755 blank cartridges are classified class "C" explosives with a minimum of susceptibility to static discharge detonation. They should be handled in a careful manner with caution against impact and shock loading.

(b) The Outdoor Firing Range Surface Danger Area is a triangle with the vertex at the gun site. Fifty meters on each side of the target shall determine the angle at the vertex (103° for test 4.4.4.1) and the maximum range shall be 300 meters.

(c) Indoor Firing Ranges shall be constructed to contain the projectile and all access shall be controlled during firing.

Custodians:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A233

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-51450A(EA)	LAUNCHER, PROJECTILE, 64 MM - M234

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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